

PROJECT BLUEPRINT: NEXUS

Real-Time Measurement & Visualization of Offline Media

Challenge Reference: Thématique 1 - The future of media in Tunisia. Moving beyond delayed, declarative reporting to build a real-time measurement ecosystem for offline media (TV, Radio, OOH).

1. Executive Summary

Nexus is a hybrid hardware-software architecture designed to bridge the data gap in offline media consumption. By combining localized edge-computing (Computer Vision), macro-level telecom infrastructure logic, and acoustic fingerprinting, we provide a unified, real-time dashboard for advertisers. This system accurately measures Out-Of-Home (OOH) impressions, precise TV/Radio tune-ins, and builds rich demographic profiles—shifting offline media into the performance-driven era of digital advertising.

2. System Architecture & Components

Component A: The OOH Hybrid Tracker (Vision + Telecom)

Measuring billboard and street-level advertising requires a scalable approach that blends ground-truth accuracy with nationwide reach.

- **The Micro (Ground Truth Calibration):** We deploy low-cost Computer Vision Edge Nodes (Raspberry Pi + Camera modules). These nodes utilize a localized, lightweight object detection model (e.g., YOLO) to instantly identify and count vehicles and pedestrians. To ensure strict privacy compliance, the hardware processes the video frames locally, discards the visual data instantly, and transmits only the numerical count and estimated demographics to our cloud infrastructure.
- **The Macro (Infrastructure Scaling):** To scale tracking across every billboard without infinite hardware costs, the system leverages cellular network architecture. We map physical billboard coordinates directly to specific mobile cell sectors. By continuously evaluating the mathematical logic of the mobile communication schemas in that sector, we calculate audience density. Specifically, we monitor real-time carrier capacity limits and analyze Carrier-to-Interference (C/I) ratios to determine the density of simultaneous communications within the target zone.

$$\text{Estimated Density } D = f(\text{Capacity}_{\text{used}}^C C/I)$$

By using the Edge Nodes to calibrate the cellular density models, we achieve highly accurate, real-time impression data for physical assets.

Component B: Ambient Audio Fingerprinting (TV & Radio)

To capture accurate TV and radio consumption without requiring user effort or massive hardware panels, we utilize a 2-second ambient audio hashing protocol.

- **Core Technology:** The system captures a brief 2-second audio sample of the user's environment. It processes this soundwave locally on the device to create a compressed mathematical hash, which is sent to our server and matched against live streams of Tunisian TV and Radio broadcasters.
- **Distribution Strategy 1 (The Trojan Horse B2C App):** A gamified consumer application. Users open the app while consuming media to unlock rewards (e.g., phone credit, promo codes, or access to live chat rooms for the show). This provides a psychological trigger (Greed, FOMO) to grant microphone access and ensure high daily active usage.
- **Distribution Strategy 2 (B2B SDK):** A lightweight SDK licensed to popular Tunisian applications (news, weather, local utilities) that passively runs the 2-second hash in the background, creating a massive, silent panel of media consumers.

Component C: Progressive User Profiling

Raw impression data is valuable, but demographic data drives advertising revenue. The B2C application integrates a progressive profiling engine.

- **Initial Onboarding:** Users provide basic data (Age, Gender) in exchange for a sign-up bonus.
- **Micro-Surveys:** When the app successfully hashes a specific TV program, it triggers a contextual micro-survey (e.g., "Do you own a car? Yes/No for 50 extra points"). Over time, this builds a deterministic, highly granular audience database that traditional media panels cannot match.

3. The "Killer Feature": Cross-Channel Attribution

The true power of Nexus lies in unifying the disconnected offline channels into a single attribution funnel. Because our system tracks both cellular OOH exposure and audio media consumption, we offer unprecedented cross-channel retargeting.

Example Scenario:

1. *2:00 PM*: The telecom macro-tracker detects the user's device in the high-density cell zone facing a major telecom billboard on Route X.
2. *8:30 PM*: The audio SDK detects the exact same user watching the telecom brand's commercial during the evening news.
3. *8:31 PM*: The Nexus platform automatically triggers a push notification to the user with a targeted promo code, converting an offline impression directly into an online sale.

4. Hackathon Demo Execution Strategy

To maximize impact with the jury, the live demonstration will focus on tangible, real-time feedback:

- **Hardware Live Feed:** The Raspberry Pi camera will be set up during the pitch, pointed at the jury/audience. The dashboard will show the "Live OOH Impressions" counter ticking up in real-time as it detects people in the room.
- **Telecom Simulation:** While the camera handles the micro-demo, the dashboard will display simulated, mathematically sound carrier capacity shifts on a map of Tunis to demonstrate the macro-scaling logic.
- **Audio Trigger:** A 5-second clip of a local Tunisian broadcast will be played from a laptop. A team member's phone (running the prototype app) will catch the audio, hash it, and instantly spike the "Live TV Viewers" graph on the dashboard, immediately popping up a dummy reward notification on the phone.